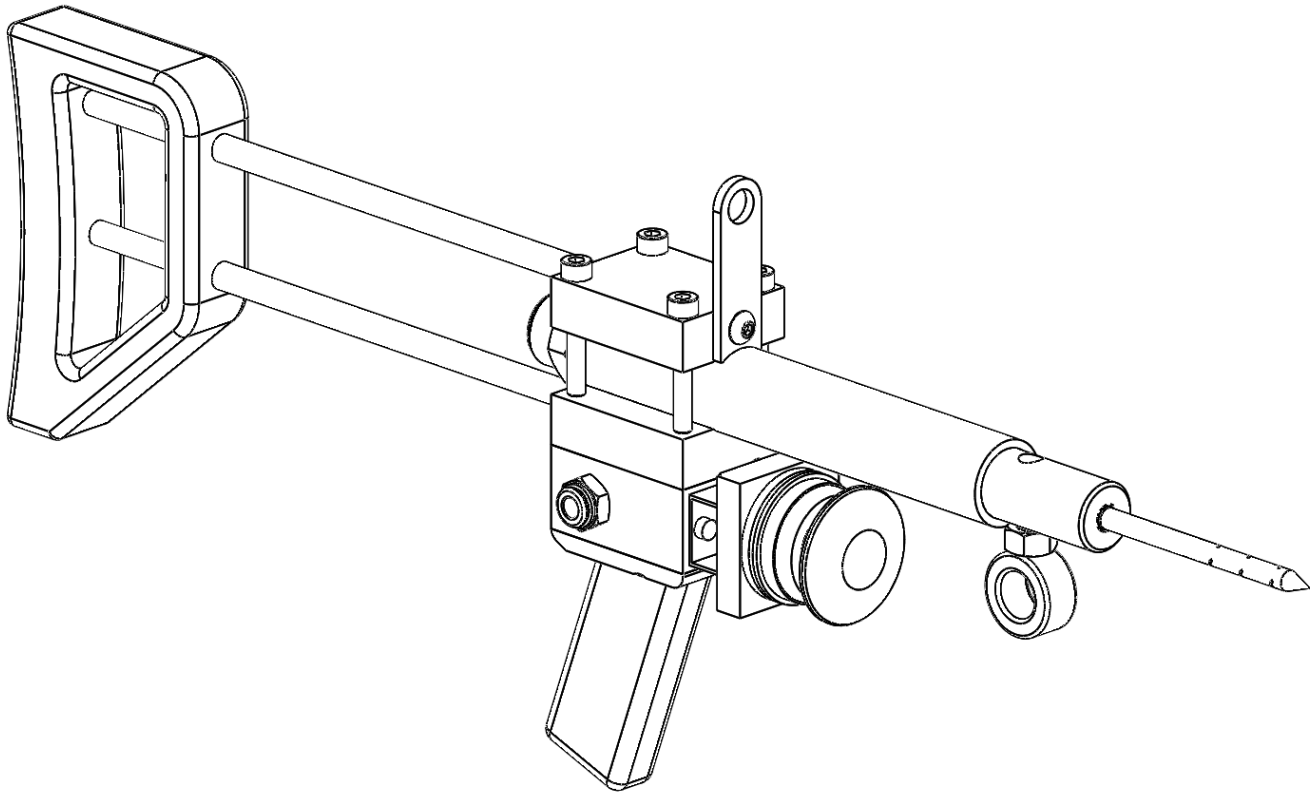


P10 Design Report - Rope Bridge DMF2



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Executive Summary

The project focused on designing and manufacturing a prototype rope bridge within a £300 budget to safeguard gibbons by avoiding road accidents. Inspired by Lego's *Build the Change* competition, the design prioritised safety, environmental harmony, and ease of installation on remote forest roads. The team was tasked with manufacturing at least three components using milling, turning, and laser-cutting techniques available in the Student Teaching Workshop (STW) and conducting a comprehensive risk assessment. Key milestones included five dedicated workshop sessions for prototype development and a design gateway review in week 5.

In the first two weeks, the team expanded the requirements outlined above into a Product Design Specification. Designs were developed for both the bridge itself and the potential installation methods and finalised using three design matrices. The team chose to focus on the bridge installation process, opting for a simple yet effective single rope design for the bridge structure due to gibbons' natural travel method of brachiation (swinging).

A pneumatic launcher was selected as the propulsion method for the projectile to launch the bridge supports into the tree since it was cost-effective, non-toxic and within STW capabilities. The design works by tying the rope to one projectile and firing it, then looping the other end through the second projectile before launching it. The trailing end can be pulled to tension and tied to a stake in the ground. This allows the rope to be retensioned throughout its service life if necessary.

Calculations were carried out to ensure the projectile had enough kinetic energy to penetrate the tree far enough to withstand the load of a family of five gibbons on the rope.

Once approved by all team members, detailed engineering drawings were created in preparation for the design review in week 5. At the design gateway, the overall design was approved with only minor revisions suggested for certain drawings.

Once these adjustments were made, the focus shifted to manufacturing a functional prototype in the Student Teaching Workshop (STW). Across the sessions, all parts were completed excluding the second projectile; however, this can be finished before the showcase in June. Assembly of the launcher was hindered by delays in delivering the manual control valve, therefore the prototype requires further testing. The modular projectile design however means that any iteration on the barbed peg will be straightforward to implement once the assembly is completed.

This report outlines the design process, engineering analysis and assembly of the prototype rope bridge launcher.

Nomenclature

A	Cross-sectional Area
CG	Centre of Gravity
D	Diameter
d	Depth of Penetration/Distance
E	Energy
F	Force
I	Second Moment of Area
K	Ratio of Outer radius to Inner Radius
KE	Kinetic Energy
m	Mass
$\dot{m}$	Mass Flow Rate
M	Maximum Bending Moment
P	Pressure
P_{atm}	Atmospheric Pressure
Q	Volumetric Flow Rate
Q_{NTP}	Normalised Volumetric Flow Rate
r	Radius
R	Gas constant
$\bar{R}$	Universal Gas Constant
SG	Specific Gravity
T	Temperature
v	Velocity
W	Work done
y	Distance from Neutral Axis
γ	Ratio of Specific Heats
ρ	Density
σ	Stress

1. Introduction

This report outlines a 10-week project with the objective to design and produce a practical solution to prevent gibbons from being killed while crossing a main road in a Malaysian rainforest, expanding on the concept of a rope bridge. The design was also required to incorporate an energy/power transmission element, which was satisfied by the pneumatic propulsion system.

The final design prioritised the installation process over the rope bridge itself. A pneumatic launcher was developed to shoot barbed pegs into trees either side of the road with rope between them. Compressed air is released by a manual control valve into the cylinder, where the gas rapidly expands and expels the projectile out. The first projectile has the rope tied on the eyebolt, the second has the rope looped through the eyebolt. The loose end is then staked into the ground allowing tension to be adjusted. The product was designed with an emphasis on the gibbons and their natural traversing methods, so the bridge consists of a single piece of rope resembling a branch.

2. Design Process/Gates

2.1 Product Design Specification (PDS)

Considering the project's objectives and functional requirements, the primary focus was structural strength, environmental impact, ease of installation within the specified surroundings, and long-term durability. These considerations are detailed in the Product Design Specification (PDS), which evolved progressively to reflect the advancements throughout the design process.

The demonstration is based on a scaled-down prototype at a 1:10 scale, except for product life, which cannot be scaled up. During testing, the projectile will be fired horizontally rather than at the scaled-down projection height of 2.5 meters (from an original 25 meters). This adjustment accounts for indoor space constraints and the challenges of controlling and predicting performance at the scaled height.

Table 1: PDS

Element	Criteria	Verification By
Customer		
Needs	Provide gibbons with a safe, natural, and easily installed road-crossing method. The infrastructure should be scalable for widespread use across the rainforest.	Animal cameras to monitor use Design Review
Safety	The design should provide the gibbons with a safe method of crossing and the installation process should not cause any injury to them.	Risk Assessment Review once tested
Competition	Other available options include cable car systems, claw machines, and various rope bridge designs.	Online research
Aesthetics	The design should harmonise with the natural environment, avoiding bright colours and incorporating natural materials.	Survey of road users
Performance	The structure must support a scaled load of 5 kg, equivalent to a family of gibbons (50 kg). The scaled rope diameter should be at least 0.2 cm to replicate the natural grip for gibbons.	Stress analysis Measuring the rope. Survey
Environment	The rope must resist high UV exposure, humidity, and temperatures up to 40°C without significant deterioration, ensuring durability for outdoor use.	Selecting an appropriate rope and evaluating its quality
Size	Aiming horizontally with 1.5 m distance, 10mm diameter	Measurement
Weight	Should be light enough to launch	Calculations and prototype tests
Ergonomics	Should accommodate gibbons' natural movement (swinging and climbing) and be easy for researchers to install.	Stress analysis
Life (based on full-scale model)		
Product Life	Aim for the design to last enough time that it is cost effective.	Testing

Service Life	Daily use of rope for 2 years	Testing
Maintenance	Aim to only recheck tension in the rope every 6 months using the threaded eyebolt, adjusting if necessary.	Testing
Producer		
Quantity	The quantity of the product must support around 2000 gibbons in South Asia's rainforest.	Ensure the product is easy to implement
Supplier	Only RS and stock STW parts	Design Review
Manufacturing constraints	Only 3 3D printed parts are allowed: Only 1 part can be made using the CNC machine, and milling, turning, and laser cutting must be used.	Design Review
Product Cost	Less than £300 for the prototype model.	BoM
Regulatory		
Product Regulations	Needs to meet the Department of Standards Malaysia's (DSM) criteria, including safety, performance, and environmental standards.	Standards and Industrial Research Institute of Malaysia
End-of-Life Disposal	Needs to follow the general environmental protection laws and adhere to best practices regarding sustainable materials.	Disposal centres and local rifle laws

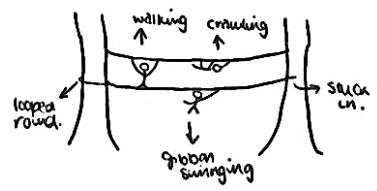
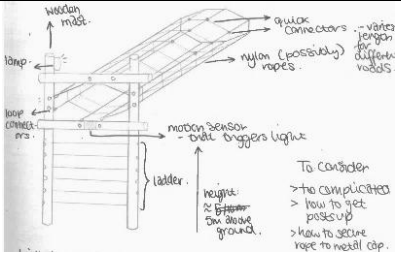
2.2 Design Decisions

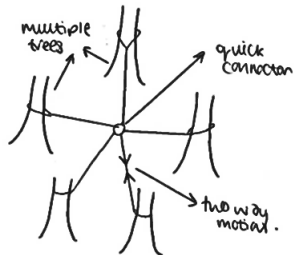
Decision matrices were used at each key design decision stage to determine the best choice among all viable options.

2.2.1 Rope Design

The system was designed with the natural movement of gibbons in mind, considering options like single, double, and multiple rope systems as well as octagonal tunnel bridges. Materials were also carefully selected, with natural fibres deemed more suitable. Research, including a study in India (Mitra, 2019), showed that gibbons prefer rope bridges over metal ones due to the latter's unfamiliar texture. Additionally, rope diameters between 2-10 cm were found to be most comfortable for gibbons, mimicking the width of branches they naturally grip.

Table 2: Initial Rope Designs

Layout	Drawing	Comments
Double Rope Bridge		Double rope opens even more movement options including handrailing, climbing, crawling upside down and swinging. More freedom and possibly suit families better with younger gibbons as this is often how they travel. We are unsure however how to attach them to the tree without causing damage.
Single Rope Bridge		Comparable to the branches the gibbons swing from in the trees but doesn't allow bipedal movement like the double or tube system. This is not actually too prevalent in the design however as gibbons only spend 10% of their time on two legs. Again, we are unsure how we would attach this to the trees
Octagonal Tunnel Bridge		Inspired by playground designs, this system features a wide tunnel allowing gibbons to walk, climb, or swing. It connects to two wooden masts with an adjustable ladder bridge between them. A sensor light enhances visibility for gibbons and alerts road users. While effective in protecting small animals from predators, similar designs have faced challenges, as animals often bypassed the tunnel by climbing on top. This complexity led to the conclusion that the system is unnecessarily intricate for the needs of the gibbons.

Multiple Rope System		This system was suggested to connect multiple trees and meet in the middle. With the quick connector in the middle, there would be the option to add/remove ropes based on the density of the trees. It would also allow gibbons to move in many directions. The rope would have to be quite taut to support the meeting in the middle, plus the weight of the connector, and therefore, it might be quite hard to swing from as there wouldn't be much give.
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These designs were then compared in the design matrix below.

Table 3: Decision matrix for rope layout

Selection Criteria	Weight (1-5)	Double Rope	Single Rope	Octagonal Tunnel	Multiple Rope
Practicality	4	4	5	4	5
Simplicity	3	4	5	1	2
Minimum Impact	3	3	4	5	2
Aesthetics	2	4	5	1	3
Safety	3	3	2	4	2
Total score		54	63	48	44

Aesthetics relates to the colour or shape of the design and how much that varies with the natural surroundings/how much it stands out. The minimum impact is how little the design affects the design has on the rainforest, e.g. drilling into the trees or putting posts into the ground.

With the added complexity of the installation process using the pneumatic launcher, as well as the data showing that a singular rope was the most popular amongst gibbons, this design was chosen. The design matrix above also proved this to be the best option.

The rope material must be water resistant, against the humid nature of the Malaysian rainforest, and protected from the relatively high heat and UV. It should have a high elastic limit so an entire family of gibbons (≈ 50 kg) can swing without causing the rope to deform plastically.

Table 4: Rope Material Comparison

Material	Advantages	Disadvantages
Nylon	Strong, can be stretched by 20% before plastic deformation – high elastic limit, resistant to shock, abrasion and oil	Not water resistant, it is easily absorbed, becoming heavy/saggy and considerably weaker.
Polyester	Can be stretched 10% before undergoing plastic deformation. Retains strength when wet	Not as strong as other rope materials.
Kevlar	Strong in tension and is resistant to cuts, abrasion, impact and most chemicals. It has a very high strength at a low mass, 2 x the strength of nylon and 5 x steel.	It is weak in compression/very rigid. Under UV radiation, webbing starts to degrade/break down, reducing the strength.

The two ropes which would be appropriate are Kevlar and Nylon. As the UV index in Malaysia can reach 11, coating the rope or replacing it regularly would need to be considered. Nylon would also need to be coated in a water-resistant material due to the humid nature of the rainforest. As Kevlar is also more expensive, Nylon was chosen for the model. This was then knotted to be secured using one of two knots: Palomar, which maintains 96% of line strength when tied, or Surgeon's, which maintains 90-95%. A Palomar knot, however, requires the rope to go through the hole twice, which is not wide enough for this, so a Surgeon's knot was chosen as there was little compromise with strength.

2.2.2 General Design Decision Matrix

The two most viable designs to install the single rope bridge were a portable elevator and a pneumatic launcher to launch rope affixation hooks into trees.

To decide which product to develop further, a decision matrix is used with the selection criteria as follows:

Table 5: Selection Criteria for General Design

Selection criteria	Weightage (1-10)	Explanation
Safety	10	With the aim to help wild gibbons, ensuring animal safety comes first when using the device is paramount.
Cost	9	With a strict budget for the prototype, keeping costs down is crucial.
Implementation	9	The designed product should be easy to implement.
Unobtrusiveness	8	Since the goal of the 'inventions' is to aid nature, the designed product mustn't disrupt the ecosystem,
Manufacture	8	Ensuring the product is easily manufactured using only the STW is essential.
Service life	5	As the Lego/New Scientist collaboration devices are intended for temporary conservation efforts, long service life is not a priority.

Table 6: Decision Matrix for installing method Design

Criteria	Weight (1-10)	Portable lift	Launcher	Scaffolding
Safety	10	8	6	7
Inexpensiveness	9	5	10	4
Implementation	9	5	8	9
Unobtrusiveness	8	6	7	1
Manufacture	8	6	7	5
Service life	6	4	8	2
Total score		284	382	247

The portable lift, towed by a pickup truck, is cost-prohibitive and requires significant time and power for installation. While the scaffolding requires no power source, the set-up process is not efficient and not stable due to uneven ground, so unsafe. The rope-launcher, however, is cost-effective, easy to build, and environmentally friendly, though it poses some safety concerns for gibbons, which are addressed in Appendix E – Risk Assessment. Therefore, through the decision matrix (Table 4), the Launcher method of installing the single rope bridge design was chosen to be developed further.

2.2.3 Rope Attachment

The design for securing the rope involved attaching it to an eyebolt for the first tree, threading it through a pulley for the second tree, and tensioning it by tying the rope around a peg at the base. The eyebolt and pulley were mounted on a plate bolted to the projectile and positioned to balance the design. After further evaluation, the eyebolt was mounted directly into the cylinder for installation and replaced the pulley for rope securing. This solution also optimally shifted the centre of gravity forward, aligning with the ideal CG, as detailed in section 3.7.

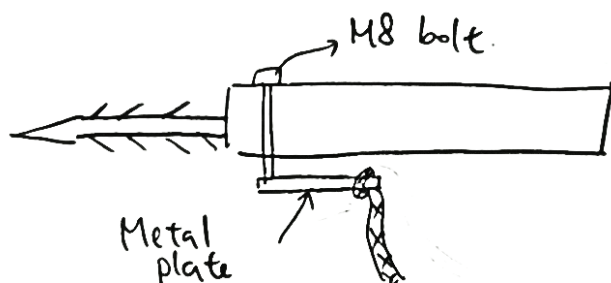


Figure 1: Initial projectile design with pulley

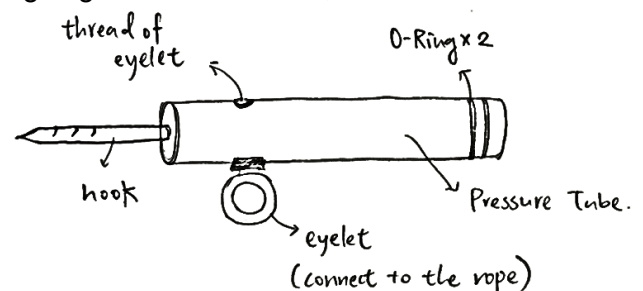


Figure 2: Final projectile design with eyebolt

2.2.4 Propulsion

To launch the projectile into the tree, four propulsion methods were researched to achieve the required projectile energy to penetrate the tree. Several ways of doing this are discussed below in Table 5.

Table 7: Propulsion comparison

Propulsion	Advantages	Disadvantages
Crossbow	Highly precise due to mechanical stability. Produces minimal noise in comparison to firearms or combustion methods. Requires no external power source	Low power/limited penetration and range. Slow reload in comparison to other methods. Often bulky for the power delivered.
Spring Launcher	Is compact while delivering decent force. Relatively cheap and simple design. Non-explosive and straightforward to build.	Power is limited by spring tension, difficult to scale for high power. Springs lose tension, limited lifespan. Less accurate compared to firearms or pneumatic systems.
Pneumatic Launcher	Can launch a wide range of projectiles, tailored for different wood. Air pressure can be tuned for desired power levels, producing adjustable forces. Easy to build, use and maintain.	Large and bulky for high-pressure designs. Slow to reload after each shot. Loud depending on pressure used, could scare nearby animals in jungle.
Railgun/Coil Launcher	Produces a high amount of power. Uses electromagnetic forces, reducing risks of combustion.	High energy consumption, complex to build, maintain, replicate on a large scale. The rails and projectile can experience extreme wear due to heat and friction.

Table 8: Decision Matrix for the Propulsion Methods

Property	Weight (1-5)	Crossbow	Spring Launcher	Pneumatic Launcher	Railgun/Coil Launcher
Power	4	3	3	5	5
Manufacturability	4	5	5	5	2
Cost	5	5	5	5	3
Size	3	2	3	5	4
Weight	2	3	4	5	3
Complexity	4	5	5	5	2
Safety	5	4	5	5	3
Power Source	4	5	5	4	2
Total score		129	139	151	92

Using the decision matrix above, it was concluded that the pneumatic launcher was the best propulsion method. It operates on the principle of rapidly releasing stored energy in the form of compressed gas to propel an object through a barrel. The key considerations for this propulsion system were the working gas, pneumatic fittings, barrel, sealing, trigger mechanism, and projectile flight characteristics.

2.2.5 Design for Manufacture

Each part underwent only essential manufacturing operations to minimise manufacturing time and costs.

The launcher is separated into 8 parts to reduce manufacturing operations. Each part has a maximum of three operations. Initial designs had the pressure cylinder and clamps as one part; however, these were separated for ease of manufacture (no longer requiring CNC).

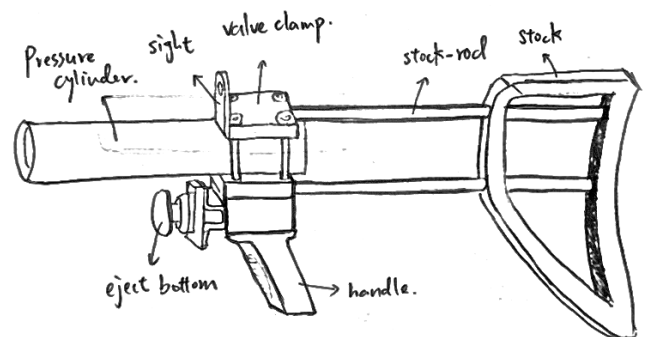


Figure 3: Final concept design for pneumatic launcher

3. Engineering analysis

3.1 Barbed Pegs

To design an effective propulsion mechanism, the energy needed for the barbed peg to penetrate the tree and remain secure under the load of gibbons using the rope bridge was calculated. The energy required to drive a peg or nail into wood can be broken down into two components: friction force and penetration force multiplied by penetration depth (Equation 1).

$$E = F \cdot d \quad (1)$$

However, the magnitude of the penetration force is negligible compared to the friction force. Therefore, calculations will be based on the friction force. For the calculations, Equation 2 about nail withdrawal force (The Engineering Toolbox, 2012) was used to estimate the force:

$$F \approx 0.95 \cdot SG^{\frac{5}{2}} \cdot D \cdot d \quad (2)$$

The dominant tree in the rainforest is the dipterocarp tree, which is hardwood with a specific gravity of 0.45-0.59. (Prasad, 2010) Using a barbed peg with $D = 6$ mm, $d = 50$ mm, and approximating $SG = 0.5$ the necessary force was calculated: $F \approx 500$ N

And the necessary projectile energy: $E = 25$ J.

The peg's 6 mm diameter, larger than typical 3-4 mm diameter nails, prevents the peg buckling on impact. Its tip is sharpened to 20° to minimise penetration force and backward-angled 45° cuts serve as barbs securing the peg in the tree. They were chosen over extruded barbs to prevent fracturing. Alternatives like ring shanks were rejected for increasing friction and required projectile energy. This calculation provides a starting point but the modular projectile design allows for iteration after testing.

3.2 Barrel

To achieve the projectile energy of 25J, proper sealing is required. Therefore, the barrel must have a surface finish of 0.8 Ra which was only achievable with a reamed hole (due to the depth and diameter of the hole required, a boring bar would have excessive deflection). This means that the maximum length of the barrel is set by the length of the reamer available in the STW (125mm).

The maximum operating pressure available was 6 bar and with a 125mm long barrel, so the required diameter of the pressure cylinder is 20.6mm, according to Equations 1 and 3.

The closest reamer size to this was 20.5mm therefore, this was the internal diameter of the pressure cylinder.

$$P = \frac{F}{A} \quad (3)$$

3.3 Working Gas

Air is the most widely available working gas requiring only a pump or compressor. There is no cost associated with the gas itself (only the compression equipment), making it ideal for budget constraints as there is a supply of air at 6 bar in the STW. It is the safest working gas as it is non-flammable and non-toxic. Both CO₂ and nitrogen were evaluated as potential working gases, as they are also common however, due to the high cost, potential toxicity and specialised equipment required in the case of nitrogen, air was selected. Air has a specific heat ratio of 1.4 and a gas constant of 287, which will be used in calculations later.

3.4 Pneumatics

To ensure the system is not restricted by the components, fittings and hose must be chosen with a maximum normalised flow rate of more than $0.0309 \text{ m}^3 \text{ s}^{-1}$ (1854 L min^{-1}). This is the maximum flow rate required to accelerate the projectile to a state with 25J of kinetic energy, calculated with Equations 4 and 5:

$$KE = \frac{1}{2}mv^2 \quad (4)$$

$$Q_{NTP} = \frac{P}{P_{NTP}} vA \quad (5)$$

The minimum diameter at any point in the system required to flow $0.0309 \text{ m}^3 \text{ s}^{-1}$ is 2.4mm, calculated with Equations 6, 7 and 8:

$$\dot{m} = P_0 A \sqrt{\frac{\gamma}{RT_0} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}} \quad (6)$$

$$Q_{NTP} = \frac{P_0}{P_{atm}} \frac{\dot{m}}{\rho} \quad (7)$$

$$P = \rho RT \quad (8)$$

4mm internal diameter pneumatic hose was used to minimise pressure drop and avoid a bottleneck in the system.

3.5 Sealing

Sufficient sealing for maximum energy transfer was achieved using two O-rings, which create an airtight seal between the projectile and the pressure cylinder walls. For optimal performance in dynamic applications, the O-rings must adhere to specific parameters: compression within 6-26%, stretch between 2-5%, and housing fill below 85%. As illustrated in Figure 4, the design meets these criteria, with detailed results in Table 10.

Table 9: O-ring Features

Dimension		Nominal Size / mm	Fit	Lower Limit/ mm	Upper Limit/ mm
Bore $\varnothing$	d_4	20.500	H7	0.000	+0.021
Projectile $\varnothing$	d_9	20.300	f7	-0.041	-0.020
Groove $\varnothing$	d_3	16.500	f7	-0.034	-0.016
Groove width	d_1	3.20		0.00	+0.25
Groove Radius	r	0.30		-0.10	+0.10
O-Ring $\varnothing$	d_1	16.00	ISO-B	-0.24	+0.24
O-Ring thickness	d_2	2.50	ISO-B	-0.09	+0.09

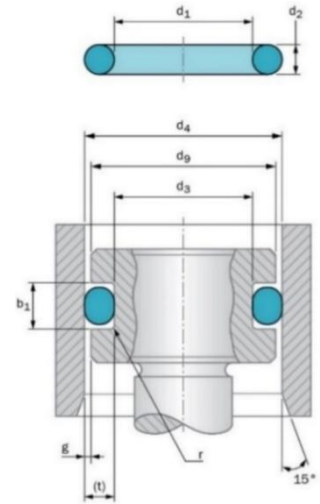


Figure 4: O-ring Specification
(Trelleborg, 2024)

Table 10: O-ring Tolerances

Result		Concentric Position		Result		Eccentric Position	
		Min.	Max.			Min.	Max.
Compression	%	13.11	21.39	Compression	%	7.50	26.11
Gap / mm	g	0.110	0.131	Gap / mm	g	0.00	0.262
Housing Fill	%	64.21	84.24				
Stretch	%	1.39	4.59				
Compression Force / mm		53	128				

3.6 Trigger Mechanism

To get a pressure of 6 bar throughout the acceleration of the projectile, the gas must be released quickly. A pneumatic valve (manual or solenoid) can be used to achieve this.

Manual control valves require no external power source; however, they do not actuate as quickly as solenoid control valves. Solenoid control valves are actuated by a solenoid coil, making them extremely fast and precise however, they are significantly more expensive. Due to budget constraints and simplicity, a manual control valve will be used.

For the design's application, a 2/2 (2 port, 2 position) manual control valve was needed. The port size of the control valve determines the flow rate. SMC have manual control valves with either G 1/4 or G 1/8 fittings. The SMC VM200 valve was chosen as it has a normalised flow rate of 6442 L min⁻¹. This gives a safety factor in a normalised flow rate of 3.47 to ensure that the valve is not a flow restriction.

3.7 Flight/Projectile Characteristics

Due to specific design requirements, such as a sharp tip for penetration, a robust eyelet for rope attachment, and a long cylinder for stable acceleration within a pneumatic launcher, the design resembles a harpoon. Harpoons generally have centres of gravity (CG) positioned between 50-65% of their total length from the tip.

SolidWorks was used to refine the dimensions of the projectile to achieve a CG within this range. Various methods were considered, such as hollowing the cylinder and adjusting its mass distribution by altering the barbed peg's length. These options were deemed impractical due to increased manufacturing complexity, cost, and safety concerns. The final design positioned the CG at 57.2%, ensuring stability and alignment with the harpoon-like function. At the projectile's high speed, rotation is minimal, making this CG placement ideal for consistent flight behaviour.

4.Failure Modes and Mitigation

4.1 Air leakage

As mentioned in section 3.5, the low surface roughness of the internal diameter of the launcher cylinder will allow for adequate sealing capabilities. In the case of the O-rings being worn out and failing to seal properly or the size of the groove being incorrect, a pack of 200 rings in varying sizes was bought. This allowed a series of trials to select the most suitable set of rings to prevent air leakage.

4.2 Structural failure of pressure cylinder

The possibility of the external cylinder failing was also explored by stress analysis. Since the ratio of the thickness to the internal radius of the launcher cylinder is less than 10, the Lamé equations (Equations 9 and 10) must be applied, with the assumption that external pressure on the cylinder is negligible.

$$\sigma_r = \frac{P}{K^2 - 1} \left(1 - \frac{r_o^2}{r^2} \right) \quad (9)$$

$$\sigma_\theta = \frac{P}{K^2 - 1} \left(1 + \frac{r_o^2}{r^2} \right) \quad (10)$$

Using the Tresca yield criterion of maximum shear stress causing yield in a ductile material, failure will first occur on the internal surface, i.e. when $r = r_i$ and can be defined by Equation 11 (Reddyhoff, 2024).

$$\tau_{max} = \frac{(\sigma_\theta - \sigma_r)}{2} \quad (11)$$

From this, the maximum shear stress found at 6 bar is 1.72 MPa . Taking the yield stress of aluminium to be $\sigma_y = 170\text{MPa}$ (Smiths Metals, 2023), a safety factor of 98.8 is produced, which is acceptable.

4.3 Structural failure of installed bridge

It was necessary to verify that the projectile would not yield under the load once lodged in the tree. The projectile was modelled as a beam of varying cross section, with a built-in end. The mass of the projectile was treated as a point load at the CG, along with the rope tension acting from the eyebolt. Based on these assumptions, the maximum stress was determined to occur at the point of contact with the tree.

To produce a safety factor, an arbitrary extreme scenario was chosen and defined as the rope forming a large angle with the axis of the projectile, creating a bigger bending moment. This angle was taken to be 70°. The maximum stress was found using Equation 12.

$$\sigma = \frac{My}{I} \quad (12)$$

From this, the largest stress was found to be $\sigma = 1.87 \text{ MPa}$. Compared with the yield stress of steel (material of the barbed pegs), which is 250 MPa (Smiths Metals, 2023), this gives a safety factor of 133, which is sufficient.

5. Manufacturing

A total of 11 parts were manufactured in the STW for the project as listed below in Table 9. Also included are a description of some of the criteria met, an overview of material selection, the chosen manufacturing method and the constraints faced whilst making the part.

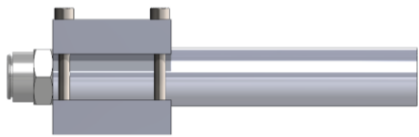
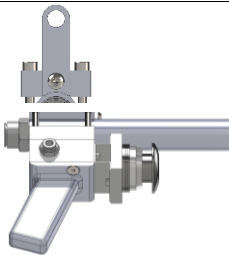
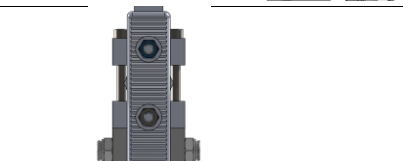
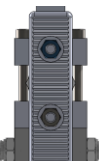
Table 11: Manufacturing and Materials Requirements

Part	Materials and Method	Requirements	Constraints
External Pressure Cylinder	Aluminium 6082 T6: Drilling, Reaming, Turning	Ra 0.8 to achieve O-ring sealing	Large internal bore required, therefore design constrained by STW reamer length
Top Clamp	Aluminium 6082 T6: Milling, Drilling	Holes must be aligned with bottom clamp, toleranced to fit with external cylinder	Part milled from large block first to allow it to be held and excess trimmed
Bottom Clamp	Aluminium 6082 T6: Milling, Drilling	Holes must be aligned with top clamp, appropriate tolerances to fit snugly with external cylinder	Part milled from large block first to allow it to be held and excess trimmed
Handle/Stock	PLA: Additive Manufacture	Ergonomic design	Holes for bolts/rods need to be hand drilled.
Stock Rod	Mild Steel: Turning	High toughness required to withstand the recoil therefore steel chosen	Small diameter meant part had to be machined at a greater length with support then cut to size
Sight	Acrylic: Laser cut	Large enough to see through	Sharp edges filed to avoid injury
Internal Pressure Cylinder	Aluminium 6082 T6: Turning, drilling, boring	Ra 1.6 to achieve O-ring sealing, lightweight aluminium to for lower energy required to accelerate	Ideally hollow but again, design constrained by STW tool sizes
Eyebolt	Mild Steel: milling, drilling	Inner edge chamfered to prevent rope fraying.	Part created as threaded feature screwed onto cylinder to avoid CNC
Barbed Pegs	Steel: turning, grinding	Peg must not deflect or fracture when embedded, high tensile strength needed	<6mm diameter deflects too much on a lathe

6. Assembly

6.1 Launcher

Table 12: Launcher assembly

Part description	Part CAD design
Mount the fitting to the threaded end of the external pressure cylinder (LNCHR.001). Place this side of the external pressure cylinder (LNCHR.001) between the top clamp (LNCHR.003) and the bottom clamp (LNCHR.004) and secure it in place using 4 x M5x45 socket head screws. Ensure that the M6 hole in both clamps is facing backwards.	
Screw the sight (LNCHR.008) to the front of the top clamp (LNCHR.003) using a M5x10 button head screw.	
Place the manual control valve (LNCHR.002) on the bottom of the bottom clamp (LNCHR.004), then place the handle (LNCHR.005) facing forwards on the bottom of the valve, and then bolt it all together using 2 x M5x40 countersunk screws.	
Screw the non-stepped side of the two stock rods into the back of the top clamp (LNCHR.003) and bottom clamp (LNCHR.004).	
Place the stock (LNCHR.007) onto the back of the stock rods and secure it with the two M8 nuts.	
Connect the small piece of the 4mm internal diameter pneumatic tube between the fitting on the left side of the valve and the fitting on the end of the external pressure cylinder (LNCHR.001). Connect the pressure tank to the fitting on the right side of the valve using a longer piece of the pneumatic hose.	

6.2 Projectile

Table 13: Projectile assembly

Place the flat side of the cylinder bracket (PRJT.002) onto the thread of the eye bolt (PRJT.003)	
Screw the eye bolt (PRJT.003) into the internal pressure cylinder (PRJT.001). Screw the barbed peg (PRJT.004) into the internal pressure cylinder (PRJT.001). Fit the O-rings into their grooves on the internal pressure cylinder.	

6.3 Rope bridge installation

To create a rope bridge, tie one end of the rope to the eye bolt of the first projectile and insert the projectile into the launcher. Check the pressure, aim, and fire it into the desired tree.

Feed the other end of the rope loosely through the eye bolt of the second projectile and follow the steps to shoot the first projectile, but do not let go of this end of the rope. After firing, tension the rope bridge by pulling the rope down and tying it to a stake in the ground.

7. Conclusion

This project's goal was to design and manufacture a prototype for implementing rope bridges into gibbon habitats, effectively bringing the 'Rope Bridge' finalist sketch from the Lego and NewScientist collaboration one step closer to reality. The team decided the best option was to engineer a device that launches rope into the trees, aiding the efforts to help wild gibbons by making it far quicker and easier to set up rope bridges in key areas, i.e., where there are busy roads to prevent the likelihood of traffic causing gibbons harm.

For the tree affixation method, a sharp barbed peg, to penetrate the bark, and an eyebolt, that the rope can be tied to, were designed. This contraption can then be launched and lodged into the trees at high speed. when the sharp barb penetrated the bark.

Since a key stipulation was that the prototype had to be demonstrated within the dimensional confines of a classroom, rope bridge length and shooting distance were scaled down to 1:10.

Other key constraints were that the launcher shouldn't produce sounds greater than 100dB (to avoid disturbing wildlife), and given the scale of the project, the prototype projectile needed to achieve a minimum flight distance of 1.5m. Having done the relevant calculations, an air pressure of 6 bars was used to achieve the required prototype performance. To achieve real flight distance for the actual product, the pneumatic values and projectile dimensions must be recalculated based on the working gas available in the jungle.

The prototype's parts were manufactured in the STW by all the team members, and assembly was completed fully before the prototype submission deadline. Despite awaiting a missing part from RS, testing was done making use of the barb's modular characteristics, determining its optimal diameter for maximum penetrative power. The prototype now awaits more rigorous testing in the summer.

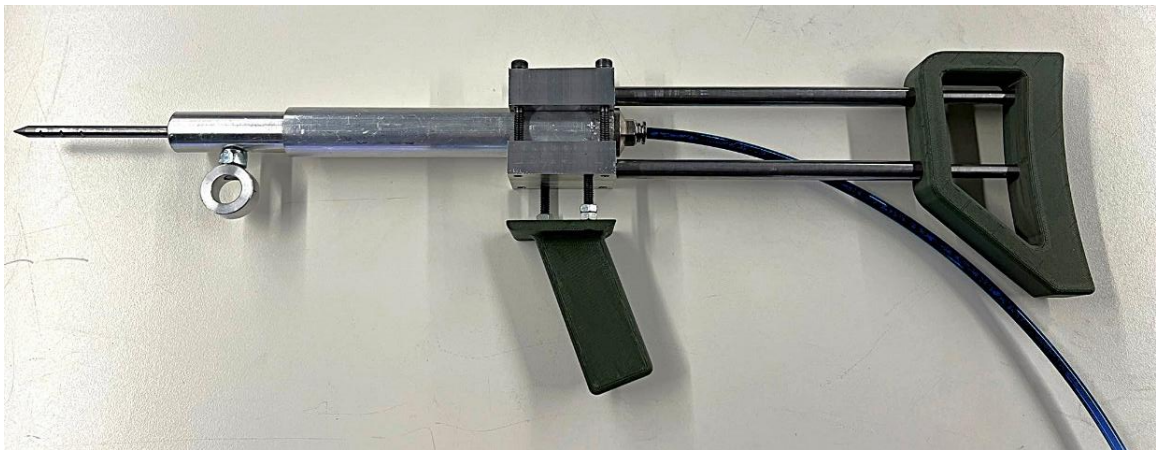


Figure 5: Prototype as of December 2024

8. References

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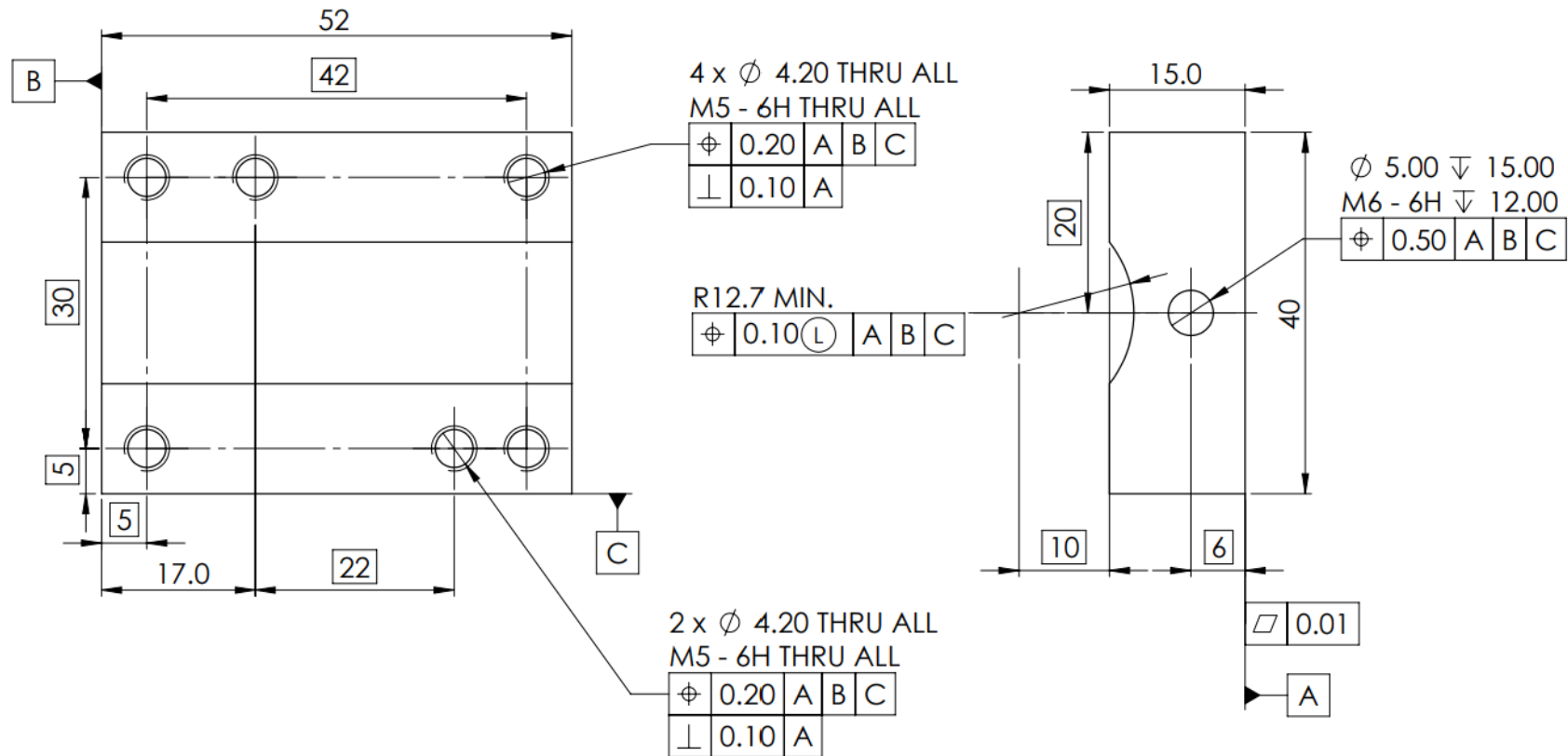
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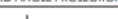
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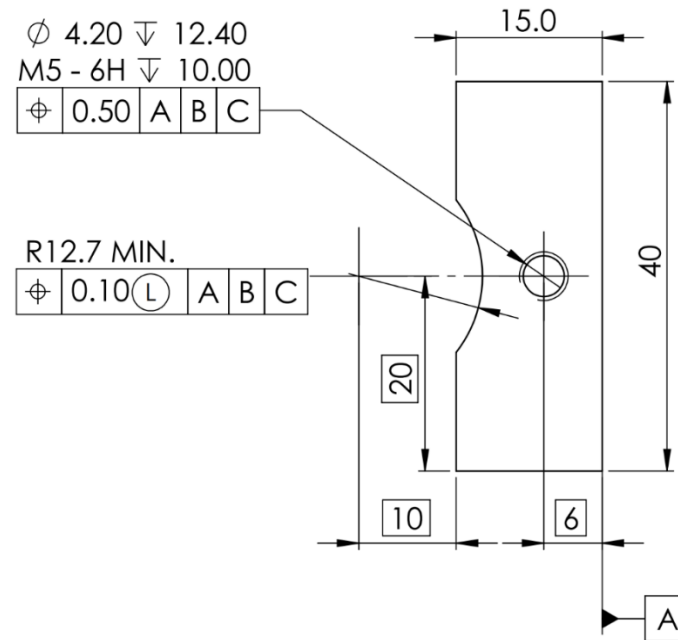
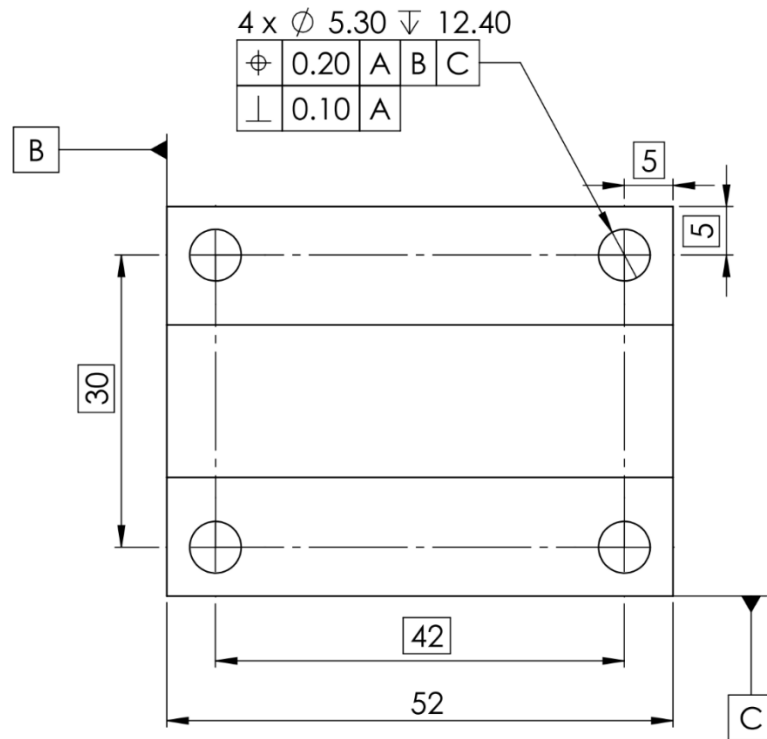
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Appendix

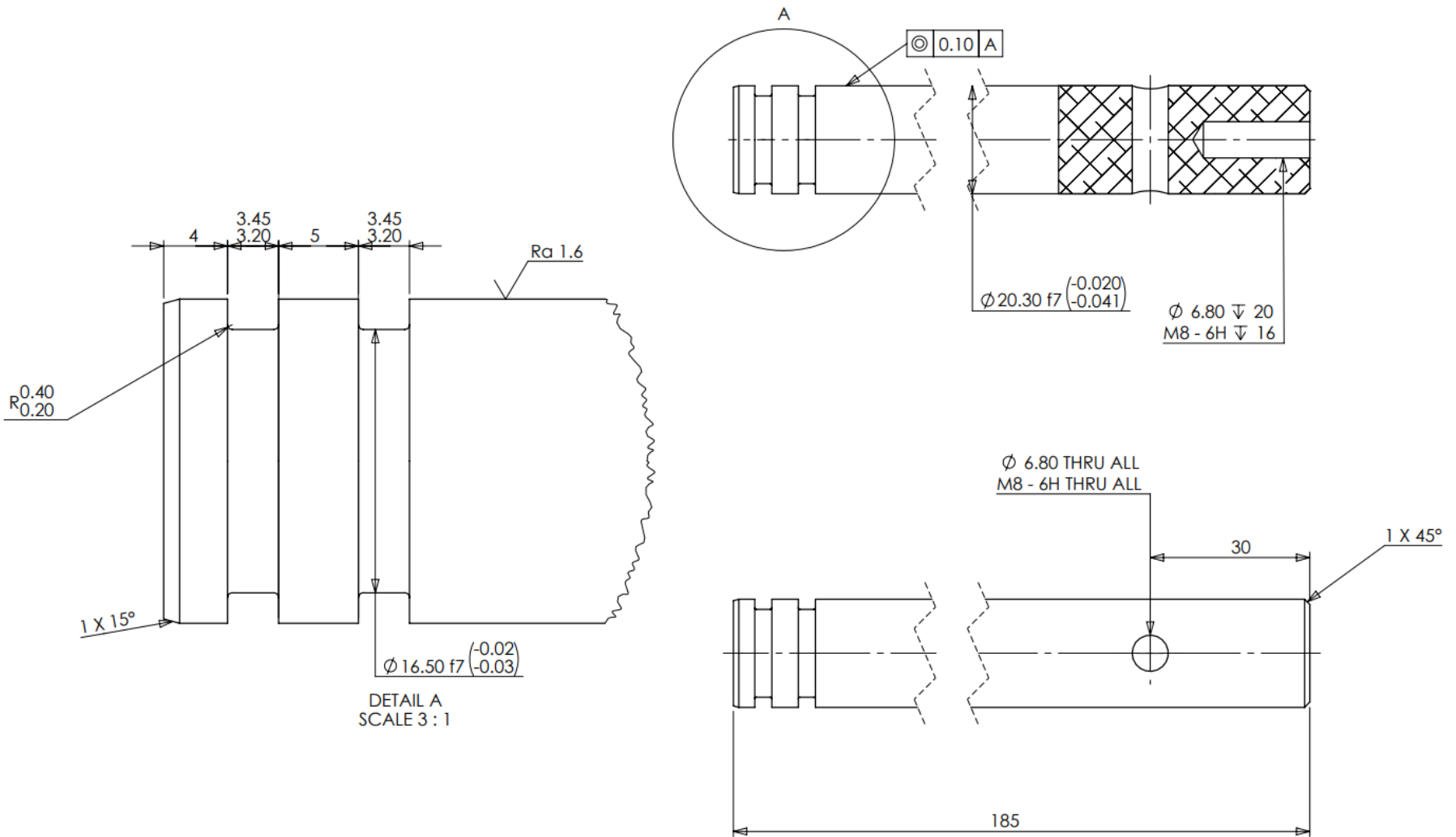
A: Part Drawings



TOLERANCES		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London Department of Mechanical Engineering	
X	± 0.5	ANGULAR $\pm 1^\circ$		ALUMINIUM 6082 T6		BOTTOM CLAMP			
X.X	± 0.1	SURFACE FINISH		ALL DIMENSIONS ARE IN MILLIMETRES					
X.XX	± 0.02	MACHINED FACES Ra 6.3							
NAME		DATE		DO NOT SCALE DRAWING		DWG No.			
DRAWN	C. JELF	25/10/24				LNCHR.004			
CHECKED	P. READ	28/10/24							
APPROVED	J. HUNGER	29/10/24		A4		SCALE 3:2			
								SHEET 1 OF 1	
								REVISION V3	



TOLERANCES		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London Department of Mechanical Engineering	
X	= ± 0.5			ALUMINIUM 6082 T6		TOP CLAMP			
X.X	= ± 0.1			ALL DIMENSIONS ARE IN MILLIMETRES					
X.XX	= ± 0.02								
		NAME	DATE	DO NOT SCALE DRAWING		DWG No. LNCHR.003			
DRAWN	C. JELF		25/10/24						
CHECKED	P. READ		28/10/24						
APPROVED	J. HUNGER		29/10/24	A4		SCALE 3:2			
								SHEET 1 OF 1	REVISION V2



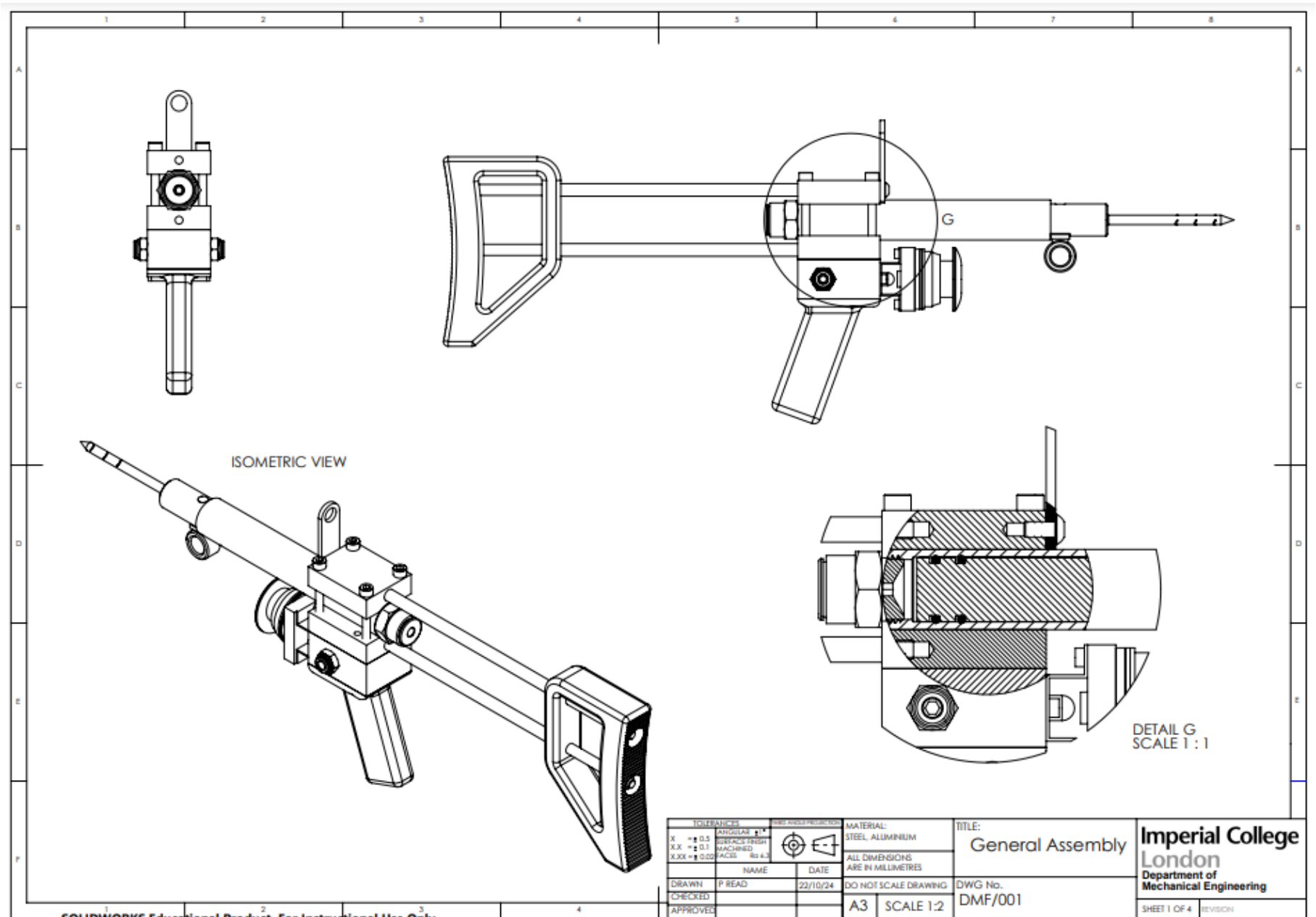
DETAIL A
SCALE 3 : 1

TOLERANCES		THIRD ANGLE PROJECTION	MATERIAL: ALUMINIUM 6082 T6		TITLE: INNER CYLINDER	Imperial College London Department of Mechanical Engineering	
X	= ± 0.5			ALL DIMENSIONS ARE IN MILLIMETRES			
X.X	= ± 0.1						
X.XX	= ± 0.02						
ANGULAR ±1°				DO NOT SCALE DRAWING	DWG No. PJRT.001		
SURFACE FINISH MACHINED FACES Ra 6.3							
DRAWN	A. JAYAKUMAR	DATE	22/10/24				
CHECKED	P. READ	DATE	28/10/24				
APPROVED	C. JELF	DATE	29/10/24	A4	SCALE 1:1		
SHEET 1 OF 1		REVISION V2					

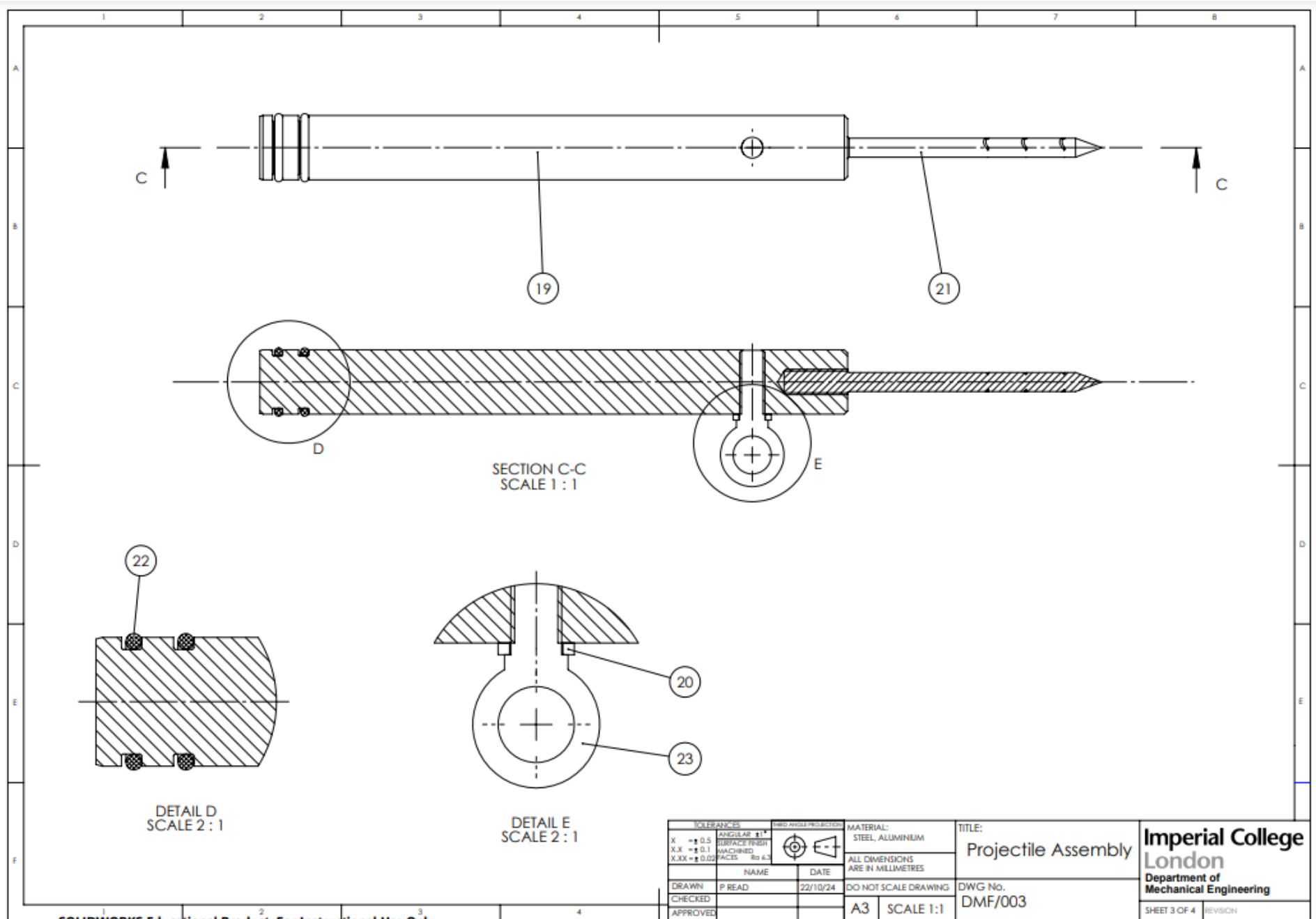
Imperial College
London
Department of
Mechanical Engineering

B: Assemblies

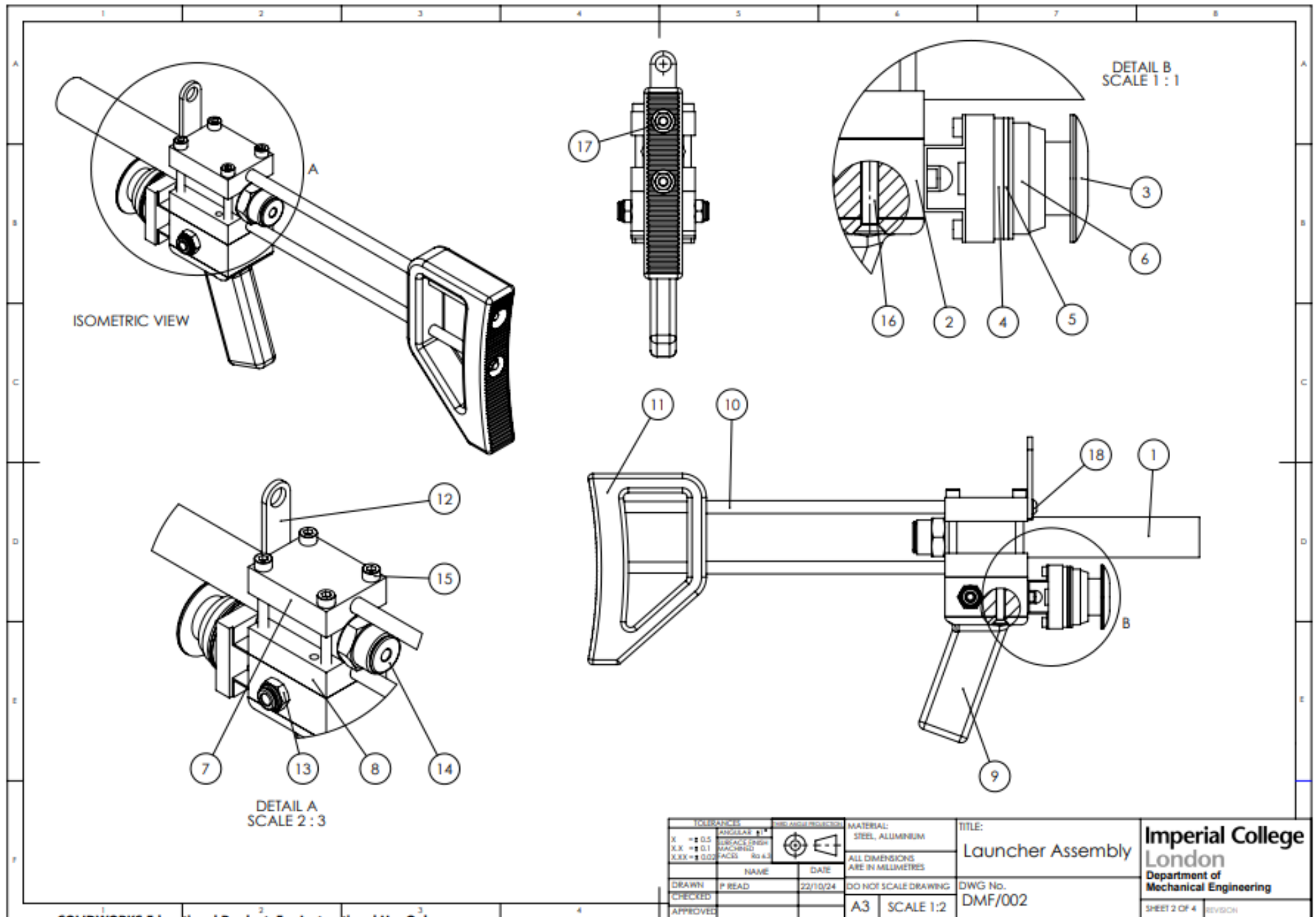
General Assembly



Projectile Sub-Assembly



Launcher Sub-assembly



TOLERANCES		MATERIAL:		TITLE:			
X	± 0.5		STEEL, ALUMINIUM	Launcher Assembly			
X.X	± 0.1						
X.X.X	± 0.02						
SURFACE FINISH MACHINED		ALL DIMENSIONS ARE IN MILLIMETRES		Imperial College London Department of Mechanical Engineering			
NAME						DATE	
DRAWN						DO NOT SCALE DRAWING	
CHECKED						DWG No.	
APPROVED			DMF/002				
		A3	SCALE 1:2	SHEET 2 OF 4			
				REVISION			

C: Bill of Materials

1	2	3	4	5	6	7	8
ITEM NO.	PART NUMBER			DESCRIPTION			QTY.
1	LNCHR.001			LAUNCHER CYLINDER			1
2	VM220-02-30BA(0_0)_BODY			SMC PUSH BUTTON			1
3	VM-30AB(0)			PNEUMATIC RELAY MANUAL CONTROL VALVE VM200 R 1/4, 1/4			1
4	VM100-30_PLATE2			BUTTON PLATE 2			1
5	VM100-30_PLATE1			BUTTON PLATE 1			2
6	VM100-30_NUT2			BUTTON NUT			1
7	LNCHR.003			UPPER CYLINDER PLATE			1
8	LNCHR.004			LOWER CYLINDER PLATE			1
9	LNCHR.005			GRIP			1
10	LNCHR.006			STOCK ROD			2
11	LNCHR.007			STOCK			1
12	LNCHR.008			SIGHT			1
13	5225K304			PNEUMATIC FITTING G1/4 TO 6MM FITTING			2
14	5225K584			PNEUMATIC FITTING G1/2 TO 6MM FITTING			1
15	ISO 4762 M5 x 45 - 45N			HEX CAP SCREW M5 X 45			4
16	ISO 10642 - M5 x 40 - 40N			HEX COUNTERSUNK SCREW M5 X 40			2
17	ISO - 4032 - M6 - W - N			HEX NUT M6			2
18	ISO 7380 - M5 x 10 - 10N			HEX BUTTON HEAD M5 X10			1
19	PRJT.001			PROJECTILE CYLINDER			1
20	PRJT.002			CURVED EYEBOLT PLATE			1
21	PRJT.004			BARBED NAIL			1
22	ORNG-16x2.5.001			NITRILE O-RING KIT			2
23	PRJT.003			M8 EYEBOLT			1

			Total:	£276.92
Description	RS Stock Number	Qty.	Price/unit	Total Price
Nylon Rope	672-992	1	£50.17	£50.17
Pneumatic Valve (2/2)	885-2127	1	£60.32	£60.32
Pressure Gauge	231-0191	1	£28.33	£28.33
Pneumatic Fitting G1/4 to 6mm pu	299-3998	2	£7.56	£15.12
Pneumatic Fitting R1/4 to 6mm pu	299-4058	2	£6.84	£13.68
Pneumatic Fitting G1/2 to 6mm pu	197-7676	1	£45.34	£45.34
O-Ring Kit	705-8958	1	£12.19	£12.19
6mm Pneumatic Hose	483-4970	1	£51.77	£51.77

TOLERANCES		HOLE ANGLES		MATERIAL:	TITLE:
X	±0.5	ANGULAR	±1°		
X.X	±0.1	SURFACE FINISH		ALL DIMENSIONS ARE IN MILLIMETRES	BoM
X.XX	±0.02	MACHINED			
		NAME	DATE	DO NOT SCALE DRAWING	DWG No. DMF/004
DRAWN	P READ	22/10/24			
CHECKED				A3	SCALE 1:1
APPROVED					

Imperial College London	
Department of Mechanical Engineering	
SHEET 4 OF 4	REVISION

D: Measurement Report on Clamps

Table 14: Top-Clamp

Dimension	Design	Position	Tolerance	Geometric Tolerance	Measurement	Feedback
Length	52 mm	X	±0.5 mm	Parallel 0.01 mm	51.79 mm	Within Tolerance
Width	40 mm	X	±0.5 mm	X	39.96 mm	Within Tolerance
Height	15.0 mm	X	±0.1 mm	X	*not machined	Ran out of time in STW
Clamping radius	R12.5 mm MIN	20 mm from C	MIN	Position (LMC) 0.1 mm	12.70 mm	Within Tolerance
Clearance Holes	4 x Ø 5.30 J12.40	5 mm from B/C	±0.2 mm	Position 0.5 mm	Max Position Deviation 0.12 mm	Within Tolerance
M5 Hole	Ø 4.20 J12.40 M5 - 6H J10.00	6 mm from B 20 mm from C	±0.2 mm	Position 0.2 mm	*not machined	Ran out of time in STW
M6 Hole	Ø 4.20 J15.00 M5 - 6H J12.00	6 mm from B 20 mm from C	±0.2 mm	Position 0.1 mm	Position Deviation 0.03 mm	Within Tolerance

Table 15: Bottom-Clamp

Dimension	Design	Position	Tolerance	Geometric Tolerance	Measurement	Feedback
Length	52 mm	X	±0.5 mm	X	52.01 mm	Within Tolerance
Width	40 mm	X	±0.5 mm	X	39.98 mm	Within Tolerance
Height	15.0 mm	X	±0.1 mm	X	*not machined	Ran out of time in STW
Clamping radius	R12.5 mm MIN	20 mm from C 10 mm from A	MIN	Position 0.1 mm	12.70mm	Within Tolerance
M5 Holes	6 x Ø 4.20 THRU ALL M5-6H THRU ALL	5 mm from B/C	±0.2 mm	Position 0.2 mm Perpendicular to A 0.1 mm	Max Position Deviation 0.08 mm	Within Tolerance
M6 Hole	Ø 4.20 J15.00 M5 - 6H J12.00	6 mm from B 20 mm from C	±0.2 mm	Position 0.1 mm	Position Deviation 0.03 mm	Within Tolerance

E: Risk Assessment

Table 163: Risk Assessment

What are the hazards?	Who might be harmed and how?	What controls can be put in to mitigate the risks?	What to do upon occurrence?	Likelihood of occurrence
Splinters or debris fly from the wood when the nail enters.	Any observers and the operators of the system	A small screen surrounds it. Anyone within a radius to wear safety goggles and stand back.	Seek medical attention. If debris has entered the eyes or skin, proceed to emergency care promptly.	Medium
The pin doesn't hit the tree/ricochet off	Anyone nearby	Screen behind. Use a large enough model to increase the probability of it hitting	Ensure no part is in the skin and respond accordingly	Low/medium
Loud noise from the compressed air, possibly leading to hearing damage and tinnitus	Produces potentially 120 dB (noise of pneumatic nail gun). Can be heard from a maximum of 18m away (in open air - this will be less due to surroundings). anyone in this radius can be affected	Use a nozzle or silencer, anyone nearby should wear ear muffers. Use a wadding jacket/absorbing material if the expenditure is an open hole instead of connected to a silencer	Seek medical care. In extreme circumstances, go to emergency care	Very low
Mishandling compressed air	Anyone facing or handling the launcher	After firing, the pressure at the tip of the launcher will be less than 30psi which is safe. Never point launcher at anyone or store once pressurised. Containment cylinder should have a high safety factor, to prevent leakage.	If ingested, can rupture stomach, intestines and lungs. Call an ambulance immediately	Low
Recoil from the pneumatic launcher	The person operating the launcher	Reduce the weight of the projectile (heavier=more recoil). Hold it firmly in the correct position against your shoulder, a stock is required for this	Injury is only likely if use is repetitive and is not held properly. Shoulders and wrists may ache so stop using and keep an eye on the pain	High (but no extreme consequences)

F: Timeline of Project

Table 174: Timeline of the project

Task	Week									
	1	2	3	4		5	6	7	8	9
Summarising design brief										
Developing a PDS										
Risk Assessment										
Initial design ideas										
Choosing design through decision matrix										
Calculations										
CAD										
Engineering Drawings										
Minor improvements to drawings										
Submit RS order, STL files (3D printing), DXF (laser cutting)										
STW sessions										
Allocate roles for report, start first draft										
Assembly of finished prototype										
Final Draft of report + poster and submission										
Logbook										

The black line represents the design Gateway.

Table 18: Individual Timelog

Time spent in hrs	Caleb Jelf	Poppy Read	Amisha Jayakumar	Joy Huang	Jakob Hunger	Total
Research and planning	12	8	10	8	15	53
Admin	2	15	10	6	4	37
Calculations and Design	30	15	22	20	22	109
CAD and Engineering drawings	25	18	15	15	15	88
Manufacturing	10	9	9	10	10	48
Poster and report	20	25	25	30	20	120
Total	99	90	91	89	86	455